

Docker Assessment

Deploy a 3-Tier Application

✓ TO DO

- **Deploy a 3-tier application using Docker Compose:**
 - Frontend
 - Backend
 - Database (official image)
- Create separate **multi-stage Dockerfiles** for the frontend and backend.
- Publish the frontend and backend images to **Docker Hub**.
- Configure Docker Compose to pull images from Docker Hub instead of building from local Dockerfiles.
- **Configure the following:**
 - Environment variables (`.env`)
 - Named volume for database persistence
 - Port mappings
 - Health checks for services
- **Implement a basic end-to-end flow:**
 - Frontend calls the backend API.
 - Backend performs a CRUD operation on the database.
 - The updated data is displayed on the frontend.

** Note: You may use GitHub or AI tools to generate a basic frontend and backend application.*

🎯 EXPECTATIONS

- The application should start successfully using: `docker compose up`
- Follow a clean and well-organized project structure.
- Include a `README.md` with:
 - Project overview
 - Setup instructions
 - Docker Hub image links
- Verify that the complete flow works: **Frontend → Backend → Database**

OUTCOME

By completing this assessment, you should be able to:

- Build optimized Docker images using multi-stage builds.
- Publish Docker images to Docker Hub.
- Deploy and manage a 3-tier application using Docker Compose.
- Configure volumes, environment variables, and health checks.
- Understand how frontend, backend, and database services work together in a containerized environment.
- Follow Docker best practices for production-ready applications.

IMPORTANT

After completing this assessment, showcase your work on **LinkedIn**! Tag **DevOps Universe** and share your experience. Documenting and sharing your journey is incredibly important for everyone in the community.